

Sample Project Documentation

Date: 29 September 2026 **Team ID:** SWTID-2026-4886

Project: AnswerBot – AI-powered FAQ and Customer Support Automation

Executive Summary

AnswerBot is a secure REST API that combines structured FAQ management with generative AI support. It provides authentication, persistent FAQ content, search, and Gemini-powered generation while keeping application logic separate from data access and external AI integration.

Team

Harshini Nair, Dharshini S, Madhumitha R — Team ID SWTID-2026-4886

Problem and Context

- Support teams repeatedly answer similar questions.
- FAQ maintenance can be manual and fragmented.
- AI assistance is useful when integrated with a controlled knowledge-management workflow.

Solution

The backend exposes REST endpoints for authentication, FAQ CRUD/search and AI generation. JWT protects private operations; bcrypt hashes passwords; Mongoose persists users and FAQs in MongoDB; Google Gemini provides generative responses.

Software Requirements

- Windows/macOS/Linux
- Node.js 18+
- npm
- Express.js
- MongoDB Atlas / MongoDB
- Mongoose
- Google Gemini SDK
- Postman or Thunder Client
- VS Code

Architecture

Layer	Components
Client	Postman / Thunder Client / future web or mobile client
Server	Express.js + Node.js
Security	JWT + bcrypt + protect middleware
Services	Authentication, FAQ, AI generation, error handling
External AI	Google Gemini
Data	MongoDB Atlas + Mongoose

Data Model

User: _id, name, email, hashed password, role, timestamps. FAQ: _id, question, answer, category, createdBy reference, timestamps. AI generation is performed dynamically by the Gemini service before a generated result is returned or used in the FAQ workflow.

API Endpoints

Method	Endpoint	Access	Purpose
POST	/api/auth/register	Public	Register user
POST	/api/auth/login	Public	Login and obtain JWT
GET/POST/PUT/DELETE	/api/faqs	Protected where applicable	FAQ management
GET	/api/faqs/search?q=...	Public	FAQ search
POST	/api/ai/answer	Protected	Generate AI answer
POST	/api/ai/generate-faq	Protected	Generate FAQ pair

Security

- Passwords are hashed with bcrypt rather than stored in plain text.
- JWT tokens are used for protected routes.
- Secrets such as database credentials and AI API keys are supplied through environment configuration.
- Errors are returned through controlled API responses rather than exposing database internals.

Testing

- Database connection was verified during project setup.
- Authentication routes were exercised.
- FAQ operations were exercised.
- AI routes were exercised after authentication/configuration was corrected.
- Formal performance benchmarking is not claimed.

Future Enhancements

- Add frontend dashboard.
- Add role-based admin moderation.
- Add feedback/rating loop for AI answers.
- Add automated test suite and formal load testing.
- Add deployment pipeline and monitoring.
- Add richer semantic retrieval using embeddings/vector search.